Terraform Init:

```
joaoribeiro@JOCAPC:~/terraform-vpc-lab/terraform-vpc-lab/vpc_lab_root$ terraform ini
t
Initializing the backend...
Initializing modules...
- vpc in ../vpc_module
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.40.0...
- Installed hashicorp/aws v6.40.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!
```

Terraform validate:

```
joaoribeiro@JOCAPC:~/terraform-vpc-lab/terraform-vpc-lab/vpc_lab_root$ terraform val
idate
Success! The configuration is valid.
```

NOTE: I redeploy everything again because I made “terraform destroy “ on the first VPC project.

Terraform plan:

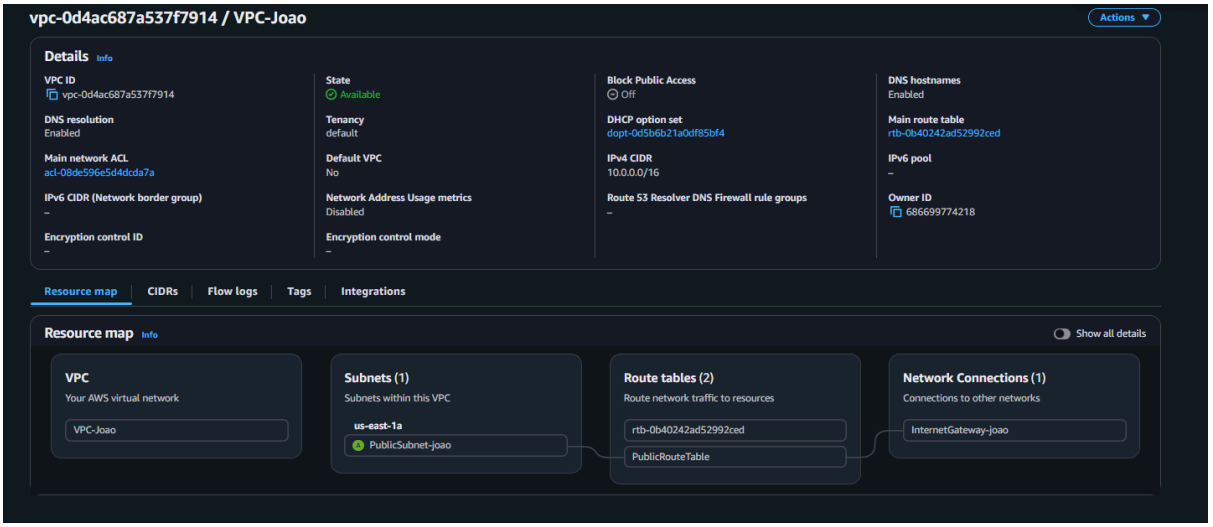
```
joaoribeiro@JOCAPC:~/terraform-vpc-lab/terraform-vpc-lab/vpc_lab_root$ terraform pla
n
module.vpc.data.aws_ami.amazon_linux: Reading...
module.vpc.data.aws_ami.amazon_linux: Read complete after 1s [id=ami-0e10497160c48e8
29]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # module.vpc.aws_instance.web_server will be created
  + resource "aws_instance" "web_server" {
      + ami                                  = "ami-0e10497160c48e829"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "Joao-useast1"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "us-east-1"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
```

```
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "WebServerInstance-joao"
        }
      + tags_all                             = {
          + "Name" = "WebServerInstance-joao"
        }
      + tenancy                              = (known after apply)
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)

      + secondary_network_interface (known after apply)
    }

  # module.vpc.aws_internet_gateway.igw will be created
  + resource "aws_internet_gateway" "igw" {
      + arn      = (known after apply)
      + id       = (known after apply)
      + owner_id = (known after apply)
      + region   = "us-east-1"
      + tags     = {
          + "Name" = "InternetGateway-joao"
        }
```

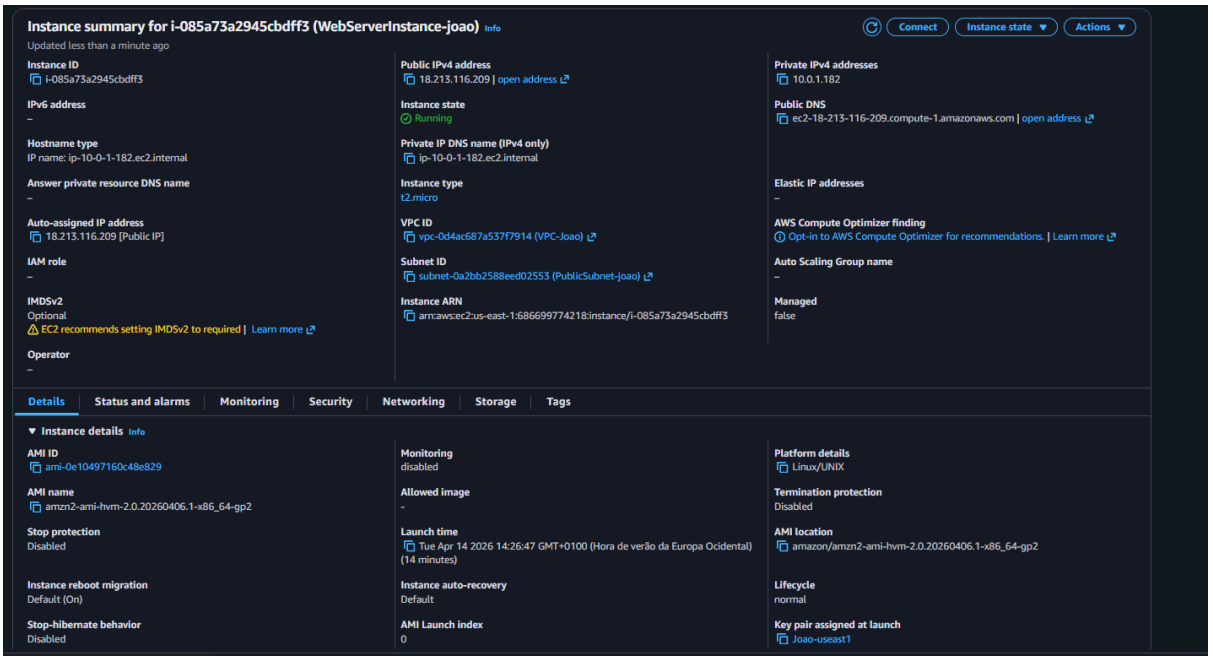
```
      + tags_all = {
          + "Name" = "InternetGateway-joao"
        }
      + vpc_id   = (known after apply)
    }

  # module.vpc.aws_route_table.public_rt will be created
  + resource "aws_route_table" "public_rt" {
      + arn             = (known after apply)
      + id              = (known after apply)
      + owner_id        = (known after apply)
      + propagating_vgws = (known after apply)
      + region          = "us-east-1"
      + route           = [
          + {
              + cidr_block                 = "0.0.0.0/0"
              + gateway_id                 = (known after apply)
                # (11 unchanged attributes hidden)
            },
        ]
      + tags            = {
          + "Name" = "PublicRouteTable"
        }
      + tags_all        = {
          + "Name" = "PublicRouteTable"
        }
      + vpc_id          = (known after apply)
    }

  # module.vpc.aws_route_table_association.public_subnet_assoc will be created
  + resource "aws_route_table_association" "public_subnet_assoc" {
      + id             = (known after apply)
      + region         = "us-east-1"
      + route_table_id = (known after apply)
      + subnet_id      = (known after apply)
    }

  # module.vpc.aws_security_group.ec2_sg will be created
  + resource "aws_security_group" "ec2_sg" {
      + arn                    = (known after apply)
      + description            = "Allow SSH access"
      + egress                 = [
          + {
              + cidr_blocks      = [
                  + "0.0.0.0/0",
                ]
              + description      = "Allow all outbound traffic"
              + from_port        = 0
              + ipv6_cidr_blocks = []
              + prefix_list_ids  = []
              + protocol         = "-1"
```

```
            + security_groups  = []
            + self             = false
            + to_port          = 0
          },
      ]
    + id                     = (known after apply)
    + ingress                = [
        + {
            + cidr_blocks      = [
                + "0.0.0.0/0",
              ]
            + description      = "SSH from anywhere (for lab purposes; restrict in
production)"
            + from_port        = 22
            + ipv6_cidr_blocks = []
            + prefix_list_ids  = []
            + protocol         = "tcp"
            + security_groups  = []
            + self             = false
            + to_port          = 22
          },
      ]
    + name                   = "EC2SecurityGroup"
    + name_prefix            = (known after apply)
    + owner_id               = (known after apply)
    + region                 = "us-east-1"
    + revoke_rules_on_delete = false
    + tags                   = {
        + "Name" = "EC2SecurityGroup_joao"
      }
    + tags_all               = {
        + "Name" = "EC2SecurityGroup_joao"
      }
    + vpc_id                 = (known after apply)
  }

# module.vpc.aws_subnet.public_subnet will be created
+ resource "aws_subnet" "public_subnet" {
    + arn                                            = (known after apply)
    + assign_ipv6_address_on_creation                = false
    + availability_zone                              = "us-east-1a"
    + availability_zone_id                           = (known after apply)
    + cidr_block                                     = "10.0.1.0/24"
    + enable_dns64                                   = false
    + enable_resource_name_dns_a_record_on_launch    = false
    + enable_resource_name_dns_aaaa_record_on_launch = false
    + id                                             = (known after apply)
    + ipv6_cidr_block                                = (known after apply)
```

VPC-Joao:



Ec2 Instance:



```
      + ipv6_cidr_block_association_id                = (known after apply)
      + ipv6_native                                   = false
      + map_public_ip_on_launch                       = true
      + owner_id                                      = (known after apply)
      + private_dns_hostname_type_on_launch           = (known after apply)
      + region                                        = "us-east-1"
      + tags                                          = {
          + "Name" = "PublicSubnet-joao"
        }
      + tags_all                                      = {
          + "Name" = "PublicSubnet-joao"
        }
      + vpc_id                                        = (known after apply)
    }

  # module.vpc.aws_vpc.my_vpc will be created
  + resource "aws_vpc" "my_vpc" {
      + arn                                  = (known after apply)
      + cidr_block                           = "10.0.0.0/16"
      + default_network_acl_id               = (known after apply)
      + default_route_table_id               = (known after apply)
      + default_security_group_id            = (known after apply)
      + dhcp_options_id                      = (known after apply)
      + enable_dns_hostnames                 = true
      + enable_dns_support                   = true
      + enable_network_address_usage_metrics = (known after apply)
      + id                                   = (known after apply)
      + instance_tenancy                     = "default"
      + ipv6_association_id                  = (known after apply)
      + ipv6_cidr_block                      = (known after apply)
      + ipv6_cidr_block_network_border_group = (known after apply)
      + main_route_table_id                  = (known after apply)
      + owner_id                             = (known after apply)
      + region                               = "us-east-1"
      + tags                                 = {
          + "Name" = "VPC-Joao"
        }
      + tags_all                             = {
          + "Name" = "VPC-Joao"
        }
    }

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + instance_public_ip = (known after apply)
```